

Legion Mark I *

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Abstract—

The Spider Bot project presents a bio-inspired robotic platform designed to support disaster response operations, particularly in post-earthquake environments where human access is unsafe or restricted. The robot mimics spider-like locomotion using a multi-legged mechanism driven by servo motors, enabling stable movement across uneven terrain, debris, and stairs. The system integrates an Arduino Nano for low-level motor and sensor control, an ESP32 module for wireless communication and video streaming, and a Raspberry Pi for advanced perception tasks such as object detection, human detection, and fire detection.

Ultrasonic sensors provide real-time obstacle detection and collision avoidance, while computer vision techniques running on the Raspberry Pi generate a 2D environmental view with estimated path distance information. This allows rescue teams to visualize the surroundings and assess navigable routes remotely. By combining mechanical design, embedded control, wireless communication, and intelligent vision-based analysis, the Spider Bot demonstrates the potential of compact, flexible robotic systems to improve safety, situational awareness, and efficiency during search-and-rescue missions in hazardous environments.

The system combines multi-legged locomotion with sensor-based obstacle avoidance and vision-based perception to enable autonomous and semi-autonomous operation. An Arduino Nano is employed for real-time control of servo motors, DC motors, and ultrasonic sensors, while an ESP32 module provides wireless communication and live video streaming. Additionally, a Raspberry Pi is integrated to handle computationally intensive tasks such as object detection, human detection, fire detection, and generation of a simplified two-dimensional environmental representation with distance estimation.

By integrating mechanical design, embedded systems, wireless communication, and computer vision techniques, the Spider Bot aims to extend human reach into dangerous environments, provide real-time situational awareness, and support informed decision-making during rescue operations. This project demonstrates how bio-inspired robotic platforms can enhance disaster response capabilities and contribute to safer and more efficient emergency management systems.

I. INTRODUCTION

Natural disasters such as earthquakes often create highly unstable and dangerous environments that restrict human access and delay rescue operations. Collapsed structures, debris, fire hazards, and narrow passages pose serious risks to rescue teams, making rapid situational assessment and victim detection extremely challenging. In such scenarios, robotic systems capable of operating in hazardous conditions can play a critical role in improving the efficiency, safety, and effectiveness of search-and-rescue missions.

Bio-inspired robotics has gained significant attention in recent years due to its ability to replicate the adaptability and stability observed in natural organisms. Spider-like robots, in particular, offer advantages such as multi-point ground contact, enhanced balance, and the ability to traverse uneven terrain where traditional wheeled or tracked robots struggle. These characteristics make legged robots well-suited for disaster environments that involve rubble, stairs, and irregular surfaces.

The Spider Bot project focuses on the design and development of a compact, intelligent, and modular robotic platform capable of navigating complex post-earthquake environments.

II. PROJECT OVERVIEW

Spider Bot project aims to develop a versatile robotic system for disaster response in post-earthquake environments where human access is limited or dangerous. The robot combines multi-legged locomotion with wheeled mobility to navigate uneven and debris-filled terrain that traditional robotic platforms struggle to traverse. This hybrid mobility approach enables stable movement across rough surfaces while maintaining efficient locomotion on flat ground. The system integrates twelve servo motors distributed across four articulated legs to achieve spider-like movement, along with four DC motors for wheeled propulsion. Ultrasonic sensors are employed for real-time obstacle detection to support safe navigation, while an ESP32-CAM module provides live video streaming and wireless communication with remote operators.

The primary objective of the project is to design and implement a reliable robotic platform capable of autonomous and semi-autonomous operation in disaster environments. An Arduino Nano microcontroller serves as the central control unit, responsible for coordinating servo motor sequencing to generate stable gait patterns, controlling DC motor motion, and processing sensor data for navigation decisions. A rechargeable battery enables untethered operation, and a power supply

module ensures stable voltage regulation across all system components.

By integrating mechanical design, embedded control systems, sensor feedback, and wireless communication, the Spider Bot extends human reach into hazardous environments and enhances situational awareness during search-and-rescue operations. The project emphasizes stability, mobility, and system reliability, validating the effectiveness of bio-inspired robotic design for navigating complex terrain and establishing a modular platform adaptable to various disaster-response scenarios

III. IMPLEMENTED HARDWARE SYSTEM

A. List of the hardwares and software Environment

The project involves integrating various electronic components and modules to create a functional automated system. Each component plays a crucial role in ensuring proper operation, control, and functionality of the overall system. Below is a detailed description of each component used in this project.

Arduino Nano: The Arduino Nano is a compact, open-source microcontroller board built around the ATmega328P processor shown in Figure 1.



Fig. 1. Arduino Nano microcontroller.

It is significantly smaller than the Arduino UNO while maintaining similar functionality, making it ideal for space constrained applications. The Arduino Nano features 14 digital input/output pins (6 of which support PWM output), 8 analog input pins, and operates at 16 MHz. It can be programmed through the Arduino Integrated Development Environment (IDE) using C/C++ based Arduino programming language. The board includes a USB connection for programming and serial communication, making it an accessible platform for prototyping and embedded system development.

ESP32-CAM Module: The ESP32-CAM is a powerful microcontroller module that combines Wi-Fi and Bluetooth connectivity with an integrated camera shown in Figure 2.

Built around the ESP32 processor, this module features an OV2640 or similar camera sensor capable of capturing images and video. It includes built-in Wi-Fi (802.11 b/g/n) and Bluetooth capabilities, allowing wireless communication and remote monitoring. The ESP32 CAM operates at 3.3V and provides numerous GPIO pins for sensor and actuator



Fig. 2. ESP32 with Camera Module microcontroller.

interfacing. Its compact size and integrated camera make it ideal for applications requiring real-time video streaming, surveillance, or computer vision tasks over wireless networks.

Ultrasonic Sensor: The ultrasonic sensor, typically the HC SR04 model, is an affordable and accurate distance measurement device with a detection range of 2 to 400 centimeters. It shown in Figure 3 operates by transmitting short ultrasonic pulses and measuring the time required for the echo to return.



Fig. 3. Ultrasonic Sonar Sensor (HC-SR04).

Based on the speed of sound and the time delay, the distance to an object can be calculated using the formula: $\text{Distance} = \text{Speed} \times \text{Time}$. The sensor features four pins: VCC (5V power supply), GND (ground), Trigger (digital output pin), and Echo (digital input pin). It provides 3-millimeter accuracy and is widely used in obstacle detection, collision avoidance, and proximity sensing applications.

Breadboard: A breadboard is a fundamental prototyping tool used for building temporary electronic circuits without soldering. It features a grid of interconnected holes where

electronic components, wires, and modules can be inserted to create circuit connections shown in Figure 4.



Fig. 4. Breadboard Medium Grid.

Breadboards are internally wired with conductive strips that allow components plugged into adjacent holes to be electrically connected. They are available in various sizes, from small mini breadboards to large prototyping boards. Breadboards are invaluable for circuit testing, design verification, and iterative development, allowing engineers to quickly modify circuits and troubleshoot issues during the prototyping phase.

Servo Motor: A servo motor is a specialized electromechanical device used for precise angular positioning and control. Unlike continuous-rotation motors, servo motors shown in Figure 5 move to specific angles based on pulse-width modulation (PWM) signals sent from a microcontroller.



Fig. 5. Servo Motor.

Commonly, servo motors provide a rotation range of 0° to 180° . They feature three connections: power (typically 5V), ground, and control signal. Servo motors are known for their high torque output despite their compact size and are extensively used in robotics, mechanical arm systems, camera pan-tilt mechanisms, and applications requiring accurate position control.

DC Motor: A DC (Direct Current) motor is an electric motor that converts electrical energy into mechanical motion through

the interaction of magnetic fields. DC motors shown in Figure 6 operate on a **simple principle**: applying voltage across the motor terminals causes the rotor to spin continuously. These motors are characterized by their simplicity, reliability, and ease of speed control through voltage regulation.



Fig. 6. DC Motor.

DC motors can be driven in either direction by reversing the polarity of the applied voltage. They are commonly used in mobile robotics, conveyor systems, fans, and any application requiring sustained rotational motion with variable speed control.

Servo Driver: The servo driver shown in Figure 7 is a specialized control module designed to manage and drive multiple servo motors simultaneously.

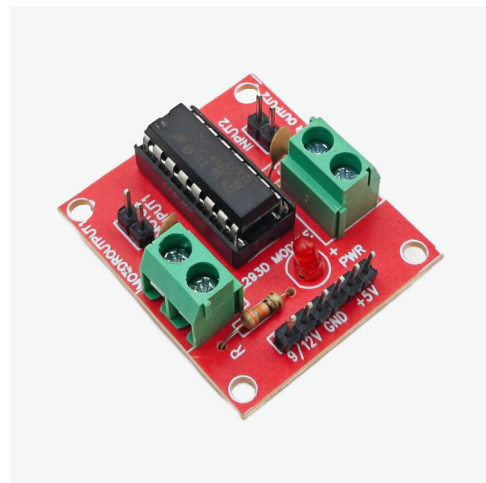


Fig. 7. L293D Motor Driver.

It provides PWM signal generation and amplification necessary to control servo motor positions precisely. Servo drivers typically feature multiple output channels, allowing independent control of several servos from a single module. They are equipped with dedicated power supply inputs to provide the required current and voltage levels that servos demand. The servo driver communicates with the microcontroller through serial protocols (I2C or SPI) or direct PWM signals, simplifying servo control logic and freeing up microcontroller resources.

Power Supply Module: The power supply module is an essential component responsible for converting and regulating electrical power for the system. It typically converts AC mains voltage to regulated DC outputs at various voltage levels (commonly 3.3V, 5V, and 12V). The module includes voltage regulation circuits, filtering capacitors, and protection mechanisms such as overcurrent protection and thermal shutdown. A reliable power supply module ensures stable operation of all components, prevents voltage fluctuations, and protects sensitive electronics from damage. Quality power supply modules feature multiple output channels to simultaneously power different components with their required voltage specifications.

Rechargeable Battery: A rechargeable battery serves as a portable power source for the system, enabling autonomous operation without dependency on external power supplies. Commonly used types include lithium-ion (Li-ion) or lithium polymer (LiPo) batteries shown in Figure 8, which offer high energy density, lightweight design, and multiple charge cycles.



Fig. 8. LiPo 3300mAh Battery.

Rechargeable batteries are characterized by their voltage output, capacity (measured in milliamp-hours or mAh), and discharge rate. These batteries provide flexibility for mobile applications and field deployment. Proper battery management, including charging circuits and discharge monitoring, is essential to maximize battery lifespan and ensure safe operation.

Jumper Wires: Jumper wires are flexible electrical conductors with connectors at both ends, used to establish temporary connections between components on a breadboard, microcontroller, and peripheral devices. They are available in various lengths and come in male-to-male, female-to-female, and male-to-female configurations to accommodate different connection requirements. Jumper wires simplify circuit prototyping by allowing quick and reversible connections without soldering. Color-coded jumper wires (red for power, black for ground, and various colors for signals) enhance circuit organization and readability, making troubleshooting and circuit modifications more efficient.

DC Motor Wheel: The DC motor wheel is a mechanical

component directly coupled to or integrated with a DC motor shaft, enabling rotational motion to be converted into linear movement. These wheels are typically made from rubber, plastic, or similar materials providing friction for traction. DC motor wheels are fundamental in mobile robotics, wheeled vehicles, and robotic platforms where locomotion is required. The wheel diameter and material composition affect the platform's speed, torque characteristics, and traction on different surfaces. Multiple DC motor wheels can be mounted on a robotic platform to enable multi-directional movement and steering control.

Raspberry Pi: The Raspberry Pi is a compact single-board computer designed for embedded computing and advanced data processing applications. It features a powerful processor, onboard memory, USB ports, HDMI output, GPIO pins, and supports operating systems such as Linux shown in Figure 9.



Fig. 9. Raspberry Pi 5 with Camera.

In this project, the Raspberry Pi is used for high-level processing tasks that require greater computational capability than a microcontroller. It processes visual data from the camera module to support advanced functionalities and communicates with the Arduino Nano for coordinated system operation. The Raspberry Pi enables the execution of complex algorithms, making it suitable for applications involving data analysis, visualization, and intelligent decision support in robotic systems.

MPU6050 Sensor :

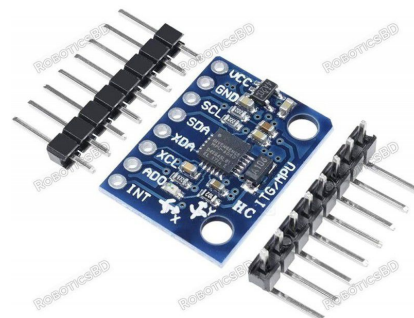


Fig. 10. MPU6050 Sensor.

The MPU6050 is a 6-axis inertial measurement unit that integrates a 3-axis accelerometer and a 3-axis gyroscope in a single chip shown in Figure 10.

It is used to measure linear acceleration and angular velocity along the X, Y, and Z axes. The accelerometer helps determine tilt and movement, while the gyroscope measures rotational motion. The sensor communicates with microcontrollers through the I²C communication protocol. It supports multiple selectable measurement ranges for flexible operation. The MPU6050 includes a built-in Digital Motion Processor for efficient motion calculations. It is compact in size and consumes low power. The sensor is widely used in robotics, drones, and motion tracking applications.

B. Figures of the implemented hardware system

The physical realization of the proposed system integrates high-level computational processing with embedded control hardware to achieve autonomous navigation and hazard detection. The core of the hardware architecture is centered around the **Arduino Nano** microcontroller, as shown in Fig. 1, which acts as the low-level execution unit for motor control and sensor data acquisition.

The hardware integration for the Legion Mark is designed to bridge high-level vision processing with multi-actuator control. As illustrated in Fig. 11, the system's electrical architecture is powered by a triple 18650 Li-ion battery configuration, providing the necessary voltage for both logic and high-current motor drive components.

a) Control and Logic Layer: The primary processing hub consists of an **Arduino Nano** interfaced with an **ESP32-CAM** module. While the ESP32-CAM handles the visual data stream for the YOLOv8-based detection (refer to Algorithm), the Arduino Nano manages the low-level signal distribution. A voltage regulator module ensures a stable power supply across these logic controllers.

b) Actuation and Power Distribution: The locomotion and manipulative capabilities are divided into two subsystems:

- **Primary Drive:** An L298N H-Bridge motor driver controls four DC gear motors, enabling the robot's movement within the mapped environment.
- **Servo Array:** A dedicated 16-channel PWM/Servo driver is utilized to manage a complex array of twelve micro-servo motors. This decoupled approach allows the Arduino Nano to control multiple degrees of freedom without taxing its onboard timers, ensuring smooth actuation for the robotic appendages.

C. Figures and Tables

a) 2D Objects Scanning Figures and Tables: The scanning system identifies objects in a 2D coordinate plane and calculates the Euclidean distance between them. As shown in Figure 12, the system successfully localized a fire source and smoke plume relative to the bot. Detailed distance metrics, including the 11.67 m separation between the fire and smoke, are summarized in Table I.

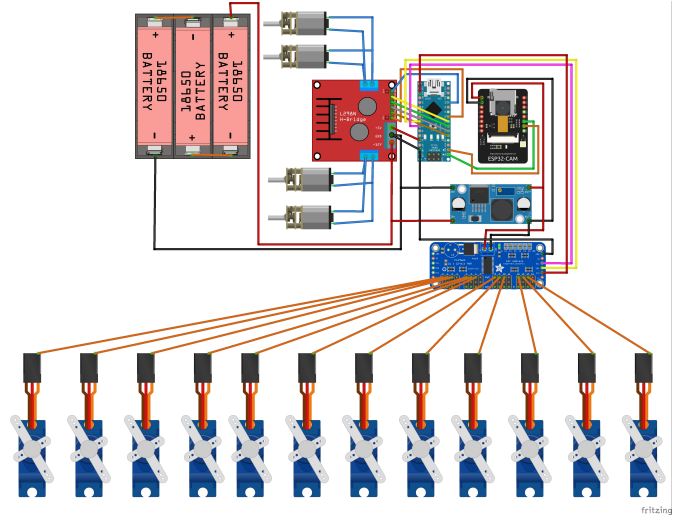


Fig. 11. Complete circuit Diagram of the Legion Mark .

TABLE I
DISTANCE MEASUREMENTS BETWEEN DETECTED OBJECTS

Source Object	Target Object and Distance (m)		
	FIRE	SMOKE	BOT
FIRE	—	11.67 m	25.23 m
SMOKE	11.67 m	—	20.45 m

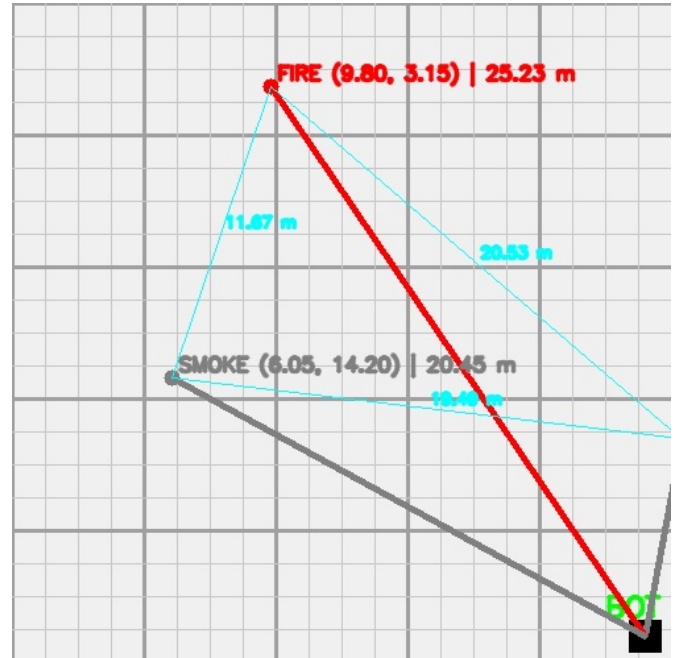


Fig. 12. 2D object Scanning Output

b) Objects Detection and Classifications: The identification and classification of environmental features are conducted through a real-time computer vision pipeline. As shown in Fig. 13, the system utilizes a bounding-box approach to isolate and classify objects within the robot's immediate vicinity. In the captured frame, the algorithm successfully identified two instances of the class *chair* with high statistical certainty. The first instance was detected with a confidence score of 0.97, measuring approximately 5.81 cm in width and 10.44 cm in height. The second instance returned a confidence score of 0.96, with dimensions of 4.56 cm width and 10.5 cm height.

The processing performance for this vision task maintained a rate of 0.57 Frames Per Second (FPS). When cross-referenced with the spatial data in Fig. 12, these visual detections allow the system to categorize obstacles located along the navigation paths between the *BOT* and identified hazards like *FIRE* and *SMOKE*. The quantitative classification data, including confidence intervals and physical dimensions, is summarized in Table II.

TABLE II
OBJECT CLASSIFICATION AND DIMENSIONAL ANALYSIS

Class	Confidence	Width (cm)	Height (cm)
chair	0.97	5.81	10.44
chair	0.96	4.56	10.5



Fig. 13. Visual detection interface showing classified objects and dimensional bounding boxes.

IV. FEATURES

The Spider Bot incorporates four major features that enable autonomous and semi-autonomous operation in challenging environments. These core capabilities are designed to enhance mobility, environmental awareness, and operational effectiveness during disaster response and exploration missions. Our Projects Feature Diagram shown on Figure 14.

Navigation & Obstacle Avoidance : The Spider Bot's four-legged design, powered by twelve servo motors, enables

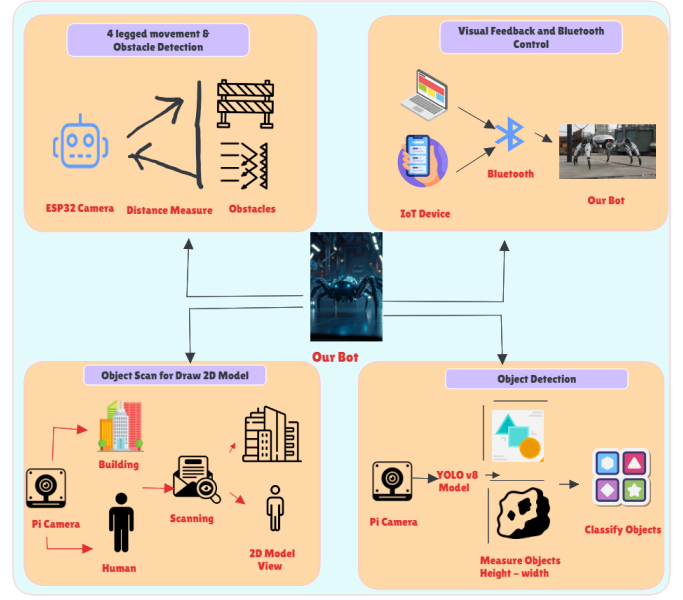


Fig. 14. Legion Features Overview

agile and stable movement across uneven and debris-filled terrains. Each leg operates independently, allowing the robot to adapt to terrain variations in real time, mimicking spider-like locomotion. Combined with an ultrasonic sensor (HC-SR04), the robot continuously measures distances to nearby obstacles and hazards, triggering audible warnings when critical thresholds are approached. This dual capability—dynamic legged movement and real-time obstacle detection—ensures safe and efficient navigation in post-disaster or challenging environments.

A. Libraries and Configuration

The following libraries are used for servo control, mathematical computation, and Arduino-based hardware interfacing:

```
#include "Legion_config.h"
#include <math.h>
#include <Arduino.h>
```

B. Servo Motor Arrangement

Each leg of the robot consists of three servo motors, resulting in a total of twelve servos. The servos are indexed using a two-dimensional array where the first index represents the leg number and the second index represents the joint.

$$\text{Servo}[4][3] \quad (1)$$

```
Servo servo[4][3];
const int servo_channels[4][3] =
{
    {0, 1, 2},
    {4, 5, 6},
    {8, 9, 10},
    {12, 13, 14}
};
```

Algorithm 1: Legion BOT Obstacle Avoidance and Navigation Logic

```

Function ObstacleAvoidanceLoop():
    while system is active do
        (pitch, roll) ← Legion::imu_update()
        // Stability Check: Anti-Tip Protection
        if |pitch| > 18° or |roll| > 18° then
            Legion::MS() // Emergency motor stop
            Legion::stand() // Restore stable posture
            continue
        // Obstacle Detection and Response
        Dobs ← GetMinDistanceToObstacle()
        if Dobs ≤ 20 cm then
            Legion::MS() // Immediate halt
            Legion::step_back(1)
            Legion::turn_left(2) // Avoid obstacle
        else if 20 cm < Dobs ≤ 50 cm then
            Legion::leg_move_speed ← ReducedSpeed
            Legion::step_forward(1)
        else
            Legion::leg_move_speed ← DefaultSpeed
            Legion::step_forward(1)

```

$$\begin{aligned}
 z_{default} &= -60 \\
 z_{up} &= -30 \\
 x_{default} &= 70 \\
 y_{default} &= 62 \\
 y_{step} &= 40
 \end{aligned} \tag{3}$$

```

float z_default = -60;
float z_up = -30;
float z_boot = z_absolute;
float x_default = 70;
float y_start = 0;
float y_step = 40;
float y_default = 62;

```

E. Speed Control Parameters

Different speed parameters are defined for smooth leg motion, body translation, and posture changes.

$$\begin{aligned}
 v_{leg} &= 8 \\
 v_{body} &= 3 \\
 v_{stand} &= 1 \\
 v_{turn} &= 4
 \end{aligned} \tag{4}$$

```

float leg_move_speed = 8;
float body_move_speed = 3;
float stand_seat_speed = 1;
float spot_turn_speed = 4;

```

C. Robot Geometrical Dimensions

The physical dimensions of the robot legs and body are defined as constant parameters (all dimensions are in millimeters).

- l_a : Coxa length
- l_b : Femur length
- l_c : Tibia length
- l_s : Distance between left and right legs
- z_{abs} : Ground reference height

$$\begin{aligned}
 l_a &= 55 \\
 l_b &= 77.5 \\
 l_c &= 27.5 \\
 l_s &= 113 \\
 z_{abs} &= -28
 \end{aligned} \tag{2}$$

D. Movement Parameters

These variables define the default posture, stepping height, and movement speed of the robot during stair climbing.

F. Turning Geometry Calculation

To enable accurate turning while climbing stairs, geometric parameters are computed using trigonometric relations.

$$\begin{aligned}
 t_a &= \sqrt{(2x_{default} + l_s)^2 + y_{step}^2} \\
 t_b &= 2(y_{start} + y_{step}) + l_s \\
 t_c &= \sqrt{(2x_{default} + l_s)^2 + (2y_{start} + y_{step} + l_s)^2}
 \end{aligned} \tag{5}$$

$$\alpha = \cos^{-1} \left(\frac{t_a^2 + t_b^2 - t_c^2}{2t_a t_b} \right) \tag{6}$$

G. Leg Position Variables

Current and target positions of each leg are stored in the following matrices:

$$\text{site}_{now}, \text{site}_{expect} \in \mathbb{R}^{4 \times 3} \tag{7}$$

```

volatile float site_now[4][3];
volatile float site_expect[4][3];
float temp_speed[4][3];

```

H. Motor Driver Pin Configuration

The DC motors responsible for body movement during stair climbing are controlled via the following pins:

```
const int motorPinA1 = 2;
const int motorPinA2 = 3;
const int motorPinB1 = 4;
const int motorPinB2 = 5;
const int enA = 6;
const int enB = 9;
```

2D Scanning & Environment Mapping: The Spider Bot leverages its live video capabilities via the ESP32-CAM module to capture real-time visual information of its surroundings. This visual data can be processed to create 2D environmental maps shown the algorithm on Algorithm 2, helping operators understand the layout of disaster zones or unknown areas. The combination of ultrasonic distance measurements and camera-based scanning enables precise mapping of obstacles and terrain features, enhancing situational awareness and supporting informed decision-making during exploration missions.

Algorithm 2: 2D Object Scanning and Spatial Mapping

Require: Input Image Frame (F), YOLO Model (M), Resolution (R)

Ensure: Generated Occupancy Map and Measurement Data

- 1: Initialize *OccupancyMap* with grid background
- 2: Set *BOT_Position* at grid coordinates ($MAP_SIZE - 20, MAP_SIZE - 20$)
- 3: $Detections \leftarrow M.predict(F)$ {Perform YOLO inference}
- 4: **for all** Object b in $Detections$ **do**
- 5: $Class \leftarrow b.getClassName()$
- 6: $Confidence \leftarrow b.getConfidence()$
- 7: **if** $Class == \text{"person"}$ **then**
- 8: $(x_1, y_1, x_2, y_2) \leftarrow b.getBoundingBox()$
- 9: $(c_x, c_y) \leftarrow \text{CalculateCentroid}(x_1, y_1, x_2, y_2)$
- 10: {Spatial Transformation}
- 11: $Dist \leftarrow$
 $\text{ComputeDistance}((c_x, c_y), BOT_Position, R)$
- 12: $(w_x, w_y) \leftarrow \text{GridToWorld}((c_x, c_y), R)$
- 13: {Visual Mapping}
- 14: Draw Circle at (c_x, c_y) on *OccupancyMap*
- 15: Draw Line from *BOT_Position* to (c_x, c_y) on *OccupancyMap*
- 16: Annotate $Dist$ on *OccupancyMap*
- 17: Record $[Class, w_x, w_y, Dist]$ into *Measurements*
- 18: **end if**
- 19: **end for**
- 20: Draw *BOT* icon at *BOT_Position* on *OccupancyMap*
- 21: **return** *OccupancyMap, Measurements*

Object Detection, Classification & Measurement: Integrated sensors and real-time video feedback allow the Spider Bot to detect and classify objects in its environment. Whether

identifying structural hazards, obstacles, or potential survivors, the system can measure distances, sizes, and positions to assist operators in planning safe navigation routes shown in Table III.

The multi-sensor approach, combining ultrasonic distance data and camera vision, creates a robust system for object recognition and measurement in dynamic and unpredictable conditions.

Remote Monitoring & Bluetooth Control : Equipped with an ESP32 module for Bluetooth connectivity, the Spider Bot can be controlled remotely through a smartphone or other mobile device. Operators can direct movement, receive live sensor data, and observe real-time video footage of the environment without needing direct line-of-sight. This wireless control system enhances operational flexibility, allowing rescue teams to monitor hazardous areas safely and coordinate response efforts efficiently.

The Legion Mark-2 utilizes an ESP32-CAM module serving a web-based asynchronous interface. The system operates on a local Access Point (IP: 192.168.4.1), providing real-time telemetry and control for a multi-legged locomotion platform.

V. CORE FUNCTIONALITIES

In the following Table III shown the web interface of the Legion Bot.

TABLE III
USER INTERFACE CONTROL MAPPING

Parameter	Functional Range
z-axis (Height)	-50 to 50 units
x-axis / y-axis	Shift/Offset Control
Leg Speed	Pulse Width Modulation (PWM) frequency
Movement Mode	Drive (Wheeled/Continuous) vs Walk (Gait-based)

After the table we can see the functional parameters will maintain the minimum 50 units distance from the camera module. Then we can see the output from the Figure 15 , so that remotely can navigate and monitoring the bots.

A. Object Detection

The video stream is processed for real-time object recognition. The probability of a detected object O belonging to class C is calculated via a lightweight convolutional neural network (CNN) such as MobileNet SSD, represented by:

$$P(C|O) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

where z represents the logit scores from the output layer.

B. Obstacle Avoidance

The "Walk Mode" integrates distance data to adjust the x, y, z coordinates of the leg gait. The safety distance d_{safe} is maintained using the following logic:

- If $d_{measured} < d_{threshold}$, trigger **Stop** or **M-Backward**.
- Velocity V is modulated by the *Movement Speed* slider, where $V_{actual} = V_{max} \cdot \frac{S}{100}$.

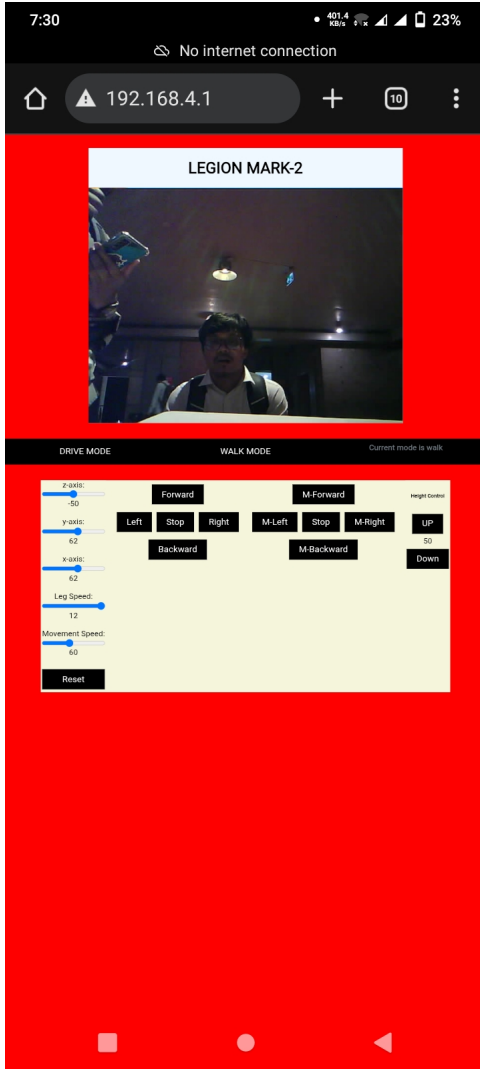


Fig. 15. Web Interface from the ESP32 Camera Module.

C. 2D Mapping and Kinematics

To achieve 2D mapping, the robot utilizes spatial coordinates. The position of each leg tip (x, y, z) relative to the body center is determined by the Inverse Kinematics (IK) transform:

$$\theta_i = f^{-1}(x, y, z)$$

where θ_i represents the servo angles for the coxae, femur, and tibia.

VI. DISCUSSION

The Spider Bot project demonstrates a bio-inspired robotic solution aimed at supporting disaster response operations, particularly in post-earthquake scenarios where human intervention is risky or impossible. By integrating multi-legged locomotion, embedded control, wireless communication, and vision-based perception, the system fulfills several functional and non-functional requirements such as terrain adaptability, remote monitoring, cost efficiency, and modular design shown in Figure 15.

A. Fulfilled Properties and Project Limitations

The proposed system successfully fulfills key properties required for disaster-response robotics, including stable locomotion on uneven terrain and stairs, real-time obstacle detection, remote visualization, and autonomous perception. The spider-like leg mechanism provides improved balance compared to wheeled robots, while the combination of ultrasonic sensing and vision-based analysis enhances situational awareness.

However, the system also has notable limitations. The robot may not perform reliably on extremely soft surfaces such as mud or loose sand, where leg slippage can occur. Additionally, heavy debris larger than the leg lifting height or stair dimensions beyond the robot's kinematic limits can prevent successful climbing. Wireless communication performance may degrade in environments with severe signal obstruction, such as deep rubble or underground areas.

B. Conflicting Requirements and Trade-off Decisions (P2)

During implementation, several conflicting requirements were encountered, particularly between performance and cost. High-speed processing for real-time object and human detection could be achieved using high-end processors, but this would significantly increase system cost and power consumption. Instead, a trade-off decision was made by using a Raspberry Pi with optimized computer vision algorithms, which provides acceptable performance while keeping the system cost-effective. This approach sacrifices some processing speed but aligns better with the project's affordability and portability requirements.

C. Alternative Designs and Solutions (P3)

One alternative solution would be a tracked or wheeled robotic platform, which could offer faster movement on flat surfaces and simpler mechanical design. However, such systems generally perform poorly on stairs and uneven debris compared to multi-legged robots. Another alternative design could involve using a single high-performance embedded system instead of distributed controllers (Arduino, ESP32, Raspberry Pi). While this could reduce system complexity and latency, it would increase cost and reduce modularity, making maintenance and upgrades more difficult.

D. Impact and Application Domain (P4)

The impact of this project extends beyond traditional computer science problems and into disaster management and emergency response. While CSE students commonly work on software-centric problems, this project required interaction with mechanical design, sensor hardware, and real-world environmental constraints. The solution addresses a real societal problem by improving safety and situational awareness for rescue teams operating in hazardous environments.

E. Standards, Documentation, and Ethical Considerations (P5)

During development, multiple hardware manuals and interfacing standards were studied, including servo motor control specifications, ultrasonic sensor timing protocols, UART

and Wi-Fi communication standards, and camera module documentation. While most implementations followed standard guidelines, certain practical adjustments were necessary due to hardware limitations and environmental noise. Ethical considerations such as user safety, non-invasive surveillance, and responsible use of collected visual data were taken into account, ensuring that the system adheres to acceptable ethical practices.

F. Stakeholder Interaction and Requirements (P6)

Feedback was obtained from peers and instructors who acted as potential stakeholders, particularly regarding usability and reliability in disaster scenarios. Their input influenced design decisions such as adding live video streaming, improving obstacle avoidance, and ensuring modular hardware connections. These requirements were incorporated to make the system more practical for real-world deployment.

G. Modularity and System Decoupling (P7)

The system is designed with a high degree of modularity. Hardware and software components are decoupled across multiple devices: the Arduino Nano handles low-level motor and sensor control, the ESP32 manages wireless communication and video streaming, and the Raspberry Pi performs computationally intensive vision tasks. Although these modules depend on each other functionally, they operate independently on separate devices, improving scalability, fault isolation, and ease of maintenance.

TABLE IV
COMPLEX ENGINEERING PROBLEM MAPPING

P1	P2	P3	P4	P5	P6	P7
✓	✓	✓	✓	✓	✓	✓

VII. CONCLUSION

The Spider Bot demonstrates a versatile and robust approach to autonomous and semi-autonomous operation in challenging environments. Its four-legged servo-driven mobility, combined with obstacle detection, real-time video feedback, and smartphone-based remote control, enables safe navigation, environmental exploration, and effective situational awareness. By integrating advanced sensing, mapping, and object recognition capabilities, the Spider Bot provides a reliable platform for disaster response, reconnaissance, and hazard assessment. Overall, this system highlights the potential of multi-functional robotic solutions to extend human capabilities, improve operational safety, and enhance efficiency in hazardous or inaccessible areas.

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equipment, workspace, and technical assistance that enabled the testing and implementation of the Spider Bot. [24]–[26]

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